Vision-Based Queue Analytics: Real-Time Queue Length and Waiting Time Estimation using Deep Learning

Abstract

Queue length and waiting time estimation are important for improving service efficiency in places such as retail stores, airports, hospitals, and public service centers. Poor queue management can lead to long waiting times and customer dissatisfaction. Traditional methods for monitoring queues often rely on manual observation or hardware-based sensors, which can be inefficient and difficult to manage in busy environments.

This project proposes a vision-based system for automatic queue length and waiting time estimation using deep learning and computer vision techniques. The system processes video input to detect and track people in a queue. A deep learning-based person detection model is used to identify individuals in each video frame, while a multi-object tracking method is applied to maintain their identities across consecutive frames.

Queue areas are defined using regions of interest (ROI). The queue length is estimated by counting the number of tracked individuals present within the defined region. Waiting time is calculated by recording the time when each person enters and exits the queue area. This approach allows the system to estimate queue-related information automatically without manual intervention.

The performance of the proposed system is evaluated using publicly available crowd datasets such as ShanghaiTech and CUHK Crowd. These datasets are used to test the accuracy of person detection and tracking in different crowd conditions. The experimental results show that the system can effectively estimate queue length and waiting time from video data.

This work demonstrates the potential of using computer vision and deep learning techniques for automated queue analysis and can serve as a foundation for further improvements in real-world queue management systems.

Introduction

Efficient queue management plays a crucial role in ensuring customer satisfaction and operational efficiency in public and commercial environments such as retail stores, airports, hospitals, and government service centers. Long waiting times and poorly managed queues often lead to customer dissatisfaction, revenue loss, and inefficient utilization of resources. Accurate estimation of queue length and waiting time is therefore essential for effective service planning and crowd management.

Problem Statement

Traditional queue monitoring techniques primarily rely on manual observation or sensor-based systems such as RFID or infrared sensors. Manual methods are labor-intensive, error-prone, and unsuitable for continuous monitoring, while sensor-based approaches require additional infrastructure and may fail to capture complex human behavior, such as people temporarily leaving or rejoining queues. These limitations highlight the need for an automated, flexible, and non-intrusive solution for queue length and waiting time estimation.

Vision-Based Queue Analytics

Recent advancements in computer vision and deep learning have enabled effective analysis of human movement in video streams. Vision-based approaches using surveillance cameras offer a cost-effective and scalable alternative, as they can utilize existing CCTV infrastructure without requiring additional sensors. By detecting individuals in video frames and tracking their movement over time, it becomes possible to estimate queue length and analyze waiting time patterns in a continuous manner. Such systems provide richer contextual information compared to traditional methods and can adapt to different environments with minimal configuration.

Challenges

Despite their advantages, vision-based queue analysis systems face several challenges. These include occlusion and overlapping of people in crowded environments, variations in camera angles and perspectives, and maintaining acceptable processing speed for near real-time performance. Additionally, changes in lighting conditions and crowd density can affect detection and tracking accuracy.

Objectives

The primary objectives of this project are:
-To develop a vision-based system for automated queue analysis
-To estimate queue length using person detection and tracking techniques
-To compute approximate waiting time based on individual movement patterns
-To visualize queue statistics through an intuitive dashboard

Applications

The proposed system can be applied in various real-world scenarios, including retail checkout counters, airport security checkpoints, hospital outpatient departments, and smart city public service centers. By providing near real-time insights into queue conditions, the system can assist authorities and service providers in optimizing resource allocation and improving user experience.

Proposed Method / Algorithm
1. Overview

The proposed system is composed of four main modules that operate sequentially to estimate queue length and waiting time from video data.

Video Input and Preprocessing:
The system accepts video streams from CCTV cameras or recorded video footage. Each frame is preprocessed through resizing and normalization to prepare it for further analysis.

Person Detection:
A deep learning-based person detection model is employed to identify individuals in each video frame. The detected bounding boxes serve as the foundation for subsequent tracking and queue analysis.

Multi-Object Tracking:
A multi-object tracking algorithm assigns unique identities to detected individuals and maintains these identities across frames. This enables consistent tracking of people as they move within and around the queue area, even in the presence of partial occlusions.

Queue Analytics:
A region of interest (ROI) corresponding to the queue area is defined. Queue length is estimated by counting the number of tracked individuals within this region. Waiting time is computed by recording the time duration between an individualâ€™s entry into the queue region and their exit after service completion.

2. Implementation Flow

The overall processing pipeline of the system is as follows:

Video Frame â†’ Person Detection â†’ Multi-Object Tracking â†’ ROI-Based Filtering â†’ Queue Statistics

To improve robustness, perspective correction techniques may be applied to handle variations in camera viewpoints. Temporal smoothing can also be used to reduce sudden fluctuations in queue length estimates.

3. Dashboard Module

The system includes a user interface for visualizing queue-related information. The dashboard displays the current queue length and estimated waiting time, providing an intuitive overview of queue conditions. This visualization aids operators and decision-makers in monitoring and managing queues more effectively.

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